



FINAL PROJECT

APPLE PRODUCT PRICING

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OBJECTIVES

01

Objective 01

Apply the full BI pipeline to a real dataset and business question.

02

Objective 02

Transform raw data into actionable insights to support decision making.

03

Objective 03

Clear recommendations backed by analysis, dashboards, and honest evaluation of limitations.

CONTEXT

This presentation outlines our comprehensive business intelligence analysis of Apple product pricing dynamics from 2020 to 2026. Developed for the Vice President of E-Commerce Channel Strategy & Commercial Operations, this project evaluates 80,000 market observations across major e-commerce platforms to uncover strategic opportunities for margin optimization, competitive positioning, and data-driven customer segmentation.



FRAME & KPI'S

01

Business question

¿How have Apple product prices evolved between 2020 and 2026, and what strategic opportunities exist to optimize margins, competitive positioning, and customer segmentation?

02

Decision maker

Vice President of E-Commerce Channel Strategy & Commercial Operations.

03

Success metrics

We are trying to generate insights that allow:

- Maximize net income
- Improve customer perception
- Increase the effectiveness of sales campaigns

04

KPIs

- Average price by category, condition.
- Variation of price vs launching price.
- Discount percentage.
- Stock availability ratio.
- Customer sentiment index.
- Impact on sell event (revenue)

DATA PREPATION & EDA

Data Cleaning & Quality

- Imputation: 92% of missing events labeled 'Regular Days'
- Anomalies: 11% 'Negative Discounts' due to scarcity/launches
- Standardization: Dropped INR; converted all analysis to USD



Business Insights

- Promotions: iPads & Watches have highest discounts (about 26%)
- Stock: Low-stock items see 10% higher discounts than average
- Growth: Avg. discounts surged from 8.7% (2020) to 26.6% (2026)
- Revenue: Macs generate the most estimated revenue (\$51B)



MODEL & EVALUATE

In this section we try to answer two related questions from our business case.

1

If we know the platform, category, condition, sale event, stock status, launch price, rating, reviews and date of a listing, can we estimate the discount we should expect?

2

Can we group listings into a few types based on price, discount, rating and review volume?

3

We use scikit-learn for both tasks.

PREDICTING DISCOUNT

The goal here is to predict the discount percentage for each listing.

01

Baseline model

has a MAE of about 14.4 percentage points, which means that always guessing the average discount is quite inaccurate.

02

Linear regression & Ridge

MAE around 6.6, so there is a clear relationship between the features and the discount. However, they still miss many cases.

03

Random forest

MAE of about 2.9, for most listings the predicted discount is within a few percentage points of the real value.

LISTING SEGMENTATION

Segmentation tries to answer what kinds of listings exist in the data

01

Segment 0

(about 10,200 listings)

High launch price (around \$1,999), low discount (~12%), mostly Mac products. These look like premium listings with little need for aggressive promotions.

02

Segment 1

(about 29,400 listings)

mid launch price (~\$744), very high discount (~39%), with the most reviews on average. Mostly iPhone. This segment behaves like a discount-driven, high-visibility group.

03

Segment 2

(about 40,300 listings)

moderate launch price (~\$862), low discount (~11%), good ratings (~4.5), mostly iPhone. This looks like a stable, mainstream segment with normal pricing.

MODEL & EVALUATE

We built a discount prediction model and a segmentation of listings.

1

For prediction, Random Forest was clearly the best option. It reduced the average error from about 14 points (baseline) to about 3 points on the test set. Cross-validation confirmed that the result is not just luck from one split.

2

For segmentation, three groups were enough to describe the marketplace: premium Mac listings, discount-heavy iPhone listings, and a larger mainstream iPhone group with moderate prices and low discounts.

3

Limitations: this is marketplace-style data, reviews are a proxy for demand but not real sales, and the segments describe listings, not individual customers. Also, conclusions should be validated before making real business decisions.

DASHBOARD

ETHICS & RESPONSIBLE USE

Data Privacy & Protection

- Zero PII: No customer personal data is collected.
- Compliance: Adheres to marketplace licensing & privacy standards.
- Secure Usage: Safe data handling protocols enforced.

Transparency & Fairness

- Unbiased Audits: Evaluated across channels (Amazon/Flipkart).
- Condition Stratification: Fair analysis for New vs Refurbished.
- Explainable AI: Clear feature importance & decision rules.

User Autonomy & Control

- Acceptable Use: Strict guidelines for commercial deployment.
- Anti-Fixing Policy: Prohibits collusion & price discrimination.
- Governance: Continuous monitoring & ethical oversight.

CONCLUSIONS

Predictive Precision

Random Forest successfully models discount behaviors with high fidelity (R^2 of 0.94), shifting strategy from blunt category averages to listing-specific accuracy.

Segment Strategy

The marketplace splits cleanly into premium Mac assets, high-volume discounted iPhones, and stable mainstream listings, demanding tailored commercial responses.

Responsible Deployment

Models must serve as decision support with strict human oversight, incorporating margin safeguards and continuous monitoring for subgroup performance drifts.



**THANK
YOU**